

Client Prompt and Behavior Tracking Table

Area	What I changed	What happened	What still needs work
Required response format	- Added ACTION: CHAT or ACTION: ANSWER. - Goal: make the server know if the agent is chatting or submitting.	- Helped separate normal messages from final answers. - Still not fully reliable because the model can format incorrectly.	- Add server-side validation. - Ignore or retry invalid formats instead of trusting the model.
Proposing / confirmation phrases	- Added I propose <figure> when an agent thinks it found the shared figure. - Added I confirm <figure> so other agents can support the proposal. - Goal: agents should agree before final answer.	- Agents often skip this step. - Sometimes one agent directly submits a figure. - Scripted phrases may confuse Gemma3:4b or make it copy patterns.	- Decide whether to keep these phrases. - Possible fix: reject ACTION: ANSWER until a valid proposal and enough confirmations exist. - Confirmations should count only if the figure is in that agent's private list.
Final answer rule	- Told agents not to answer until there is enough support. - Goal: prevent early final answers.	- Useful rule, but the model does not always follow it. - One agent may still answer too early.	- Put the rule in server code. - Server should block final answer before the proposal/confirmation condition is met.
Conversation history	- Reduced history sent to each Jetson. - Example: last 7 public messages instead of full history.	- Less history should reduce confusion and repeated text. - Too much history seemed to increase looping.	- Test if 7 messages is enough for 3-5 Jetsons. - Adjust by topology and number of agents.
Prompt examples	- Reduced exact examples and step-by-step scripts. - Goal: avoid making the model copy the prompt.	- This is better for Gemma3:4b. - Too many examples made agents rigid or repetitive.	- Find balance: enough guidance, not too much scripting. - Keep examples only if they clearly improve behavior.
Message length	- Told agents to keep messages short and relevant.	- Shorter messages make logs easier to read. - They also reduce repeated text in history.	- Limit output length in model parameters. - Consider trimming or rejecting very long replies.
Model parameters in client	- Looked at temperature, top_p, repeat_penalty, and output length.	- These affect stability, randomness, and looping. - Low temperature can be stable but repetitive.	- Run comparison trials. - Test temperature 0.0 vs 0.1, repeat penalty, top_p, and max output length.
Looping behavior	- Reduced prompt scripting. - Reduced history. - Added/planned max rounds and manual stop.	- Still a main unresolved issue. - Agents can repeat proposals, confirmations, or similar chat messages.	- Add stronger loop detection. - Stop or reset trial after repeated rounds. - Continue testing prompt and parameters.
Small-model limitation	- Rewrote prompt for Gemma3:4b instead of assuming strong instruction-following.	- Main lesson: the model cannot reliably follow complex multi-step rules.	- Keep prompt simple. - Move critical logic to server code: valid figures, proposal before answer, max rounds.
Client reliability	- Considered restart/reload support when Jetsons stop responding or fail to reconnect.	- Some failures are networking/service issues, not prompt issues.	- Test server menu restart option. - Make client reconnection reliable after each trial.
Trial reset	- Noticed every trial must start clean. - No old history, proposal, confirmations, or figures should remain.	- Old state can contaminate a new trial and cause strange behavior or loops.	- Confirm full reset on server and client side. - New trial should start with fresh state only.

Main takeaway: keep the client prompt simple, and enforce critical rules in server code whenever possible.